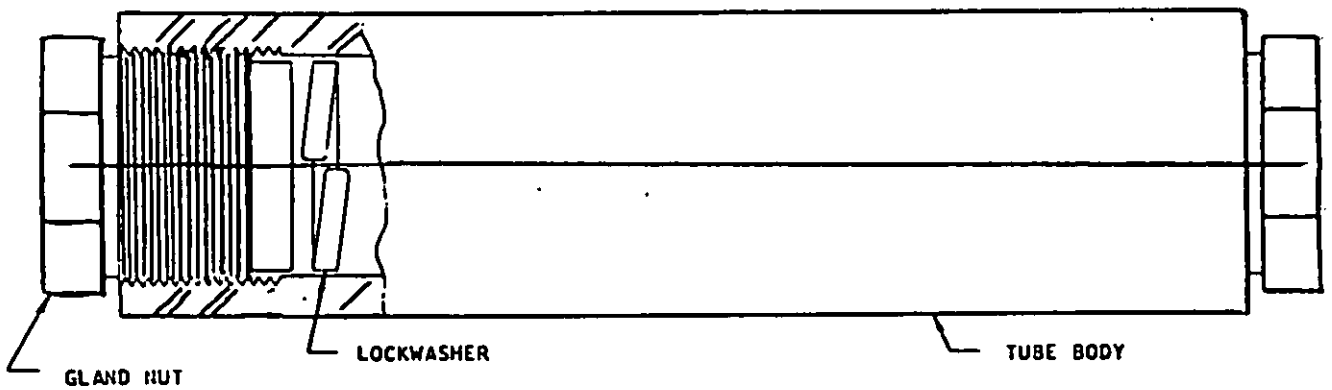
INCH-POUND
MIL-S-24235/1E(SH)
21 August 1989
SUPERSEDING
MIL-S-24235/1D(SH)
25 October 1977

MILITARY SPECIFICATION SHEET

STUFFING TUBES, METAL, AND PACKING ASSEMBLIES FOR ELECTRIC CABLES, BULKHEAD, PRESSUREPROOF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24235.



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FIGURE 1. Stuffing tube assembly. (01)

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances shall be as follows:
   Decimals - plus or minus 0.005 inch.
   Fractions - plus or minus 1/64 inch.
   Angular dimensions - plus or minus 30 minutes.

AMSC N/A FSC 5975

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

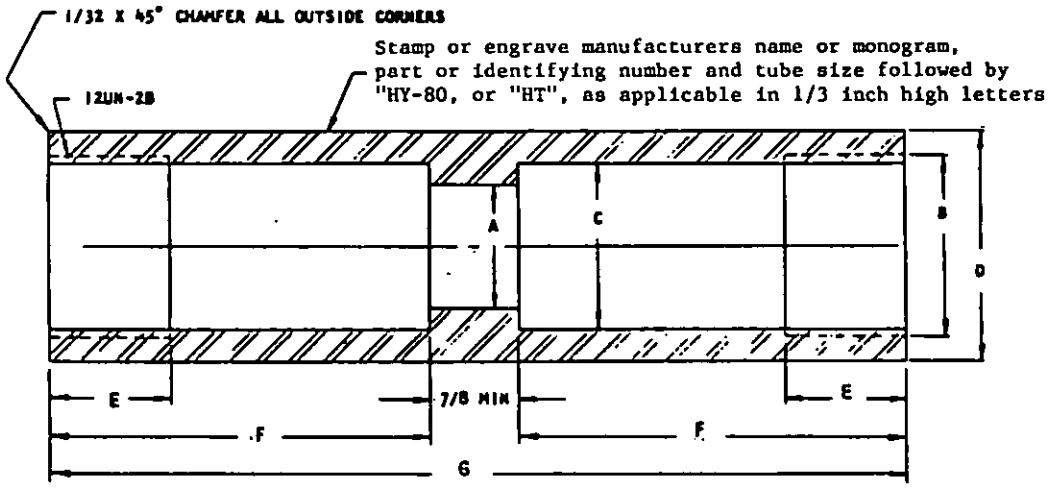
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TABLE I. Sizes and part number - HY-80.

Tube size	Symbol number	Assembly part number	Part or identifying numbers of components		
			Tube body 1 required	Gland nut 2 required	Lockwasher 2 required
1	2405 HY-80	M24235/1-001	M24235/1-010	M24235/1-019	M24235/1-028
2	2406 HY-80	M24235/1-002	M24235/1-011	M24235/1-020	M24235/1-029
3	2407 HY-80	M24235/1-003	M24235/1-012	M24235/1-021	M24235/1-030
4	2408 HY-80	M24235/1-004	M24235/1-013	M24235/1-022	M24235/1-031
5	2409 HY-80	M24235/1-005	M24235/1-014	M24235/1-023	M24235/1-032
6	2410 HY-80	M24235/1-006	M24235/1-015	M24235/1-024	M24235/1-033
7	2411 HY-80	M24235/1-007	M24235/1-016	M24235/1-025	M24235/1-034
8	2412 HY-80	M24235/1-008	M24235/1-017	M24235/1-026	M24235/1-035
9	2413 HY-80	M24235/1-009	M24235/1-018	M24235/1-027	M24235/1-036

TABLE II. Sizes and part or identifying numbers - HT.

Tube size	Symbol number	Assembly part number	Part or identifying numbers of components		
			Tube body 1 required	Gland nut 2 required	Lockwasher 2 required
1	2405 HT	M24235/1-101	M24235/1-110	M24235/1-119	M24235/1-128
2	2406 HT	M24235/1-102	M24235/1-111	M24235/1-120	M24235/1-129
3	2407 HT	M24235/1-103	M24235/1-112	M24235/1-121	M24235/1-130
4	2408 HT	M24235/1-104	M24235/1-113	M24235/1-122	M24235/1-131
5	2409 HT	M24235/1-105	M24235/1-114	M24235/1-123	M24235/1-132
6	2410 HT	M24235/1-106	M24235/1-115	M24235/1-124	M24235/1-133
7	2411 HT	M24235/1-107	M24235/1-116	M24235/1-125	M24235/1-134
8	2412 HT	M24235/1-108	M24235/1-117	M24235/1-126	M24235/1-135
9	2413 HT	M24235/1-109	M24235/1-118	M24235/1-127	M24235/1-136



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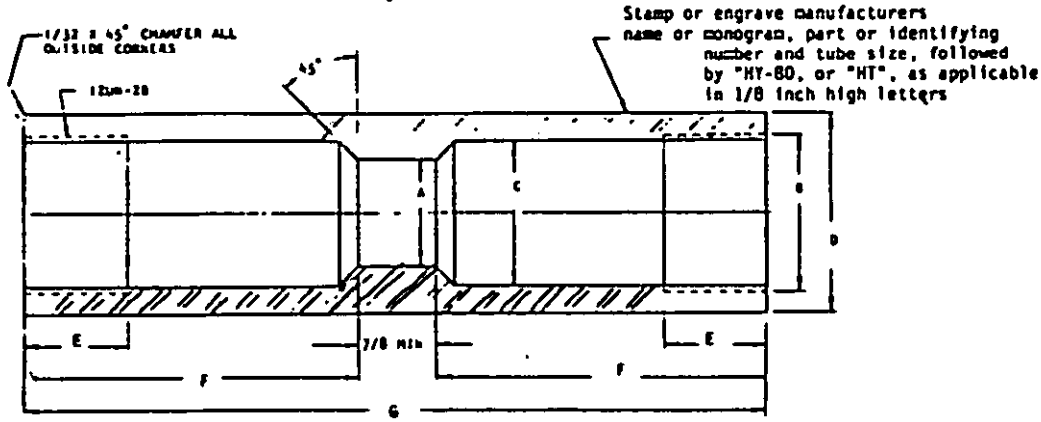
FIGURE 2. Tube body - sizes 1, 2 and 3. (02)

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TABLE III. Tube body dimensions.

Tube size	Part or identifying number HY-80	Part or identifying number HT	A DIA	B THDS	C DIA	D DIA	E 1/	F	G
1	M24235/1-010	M24235/1-110	0.609	1.125	1.017	1-1/2	15/16	2-3/8	5-5/8
2	M24235/1-011	M24235/1-111	.812	1.312	1.204	1-11/16	15/16	2-11/16	6-1/4
3	M24235/1-012	M24235/1-112	1.062	1.625	1.517	2-1/8	15/16	3-1/8	7-1/8

1/ Tolerance shall be minus 0, plus 4 threads.



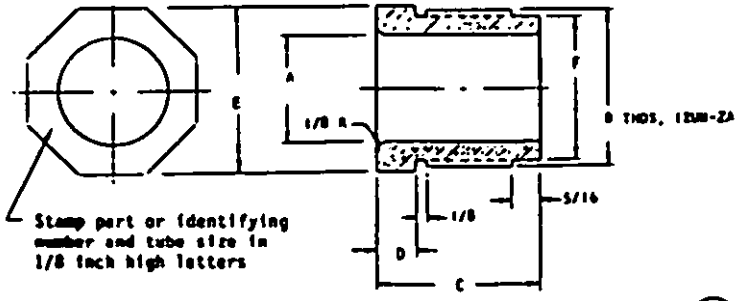
SH 9166B

FIGURE 3. Tube body - sizes 4 through 9. (03)

TABLE IV. Tube body dimensions.

Tube size	Part or identifying number HY-80	Part or identifying number HT	A DIA	B THDS	C DIA	D DIA	E 1/	F	G
4	M24235/1-013	M24235/1-113	1.265	1.812	1.704	2-5/16	1-1/4	3-11/16	8-1/2
5	M24235/1-014	M24235/1-114	1.437	2.062	1.954	2-11/16	1-3/8	4-9/16	10
6	M24235/1-015	M24235/1-115	1.625	2.312	2.204	2-15/16	1-1/2	4-7/8	10-5/8
7	M24235/1-016	M24235/1-116	1.875	2.688	2.578	3-5/16	1-11/16	5-1/4	11-3/8
8	M24235/1-017	M24235/1-117	2.187	3.000	2.882	3-15/16	1-11/16	5-3/8	11-5/8
9	M24235/1-018	M24235/1-118	2.312	3.125	3.077	3-15/16	1-3/4	5-1/2	11-7/8

1/ Tolerance shall be minus 0, plus 4 threads.



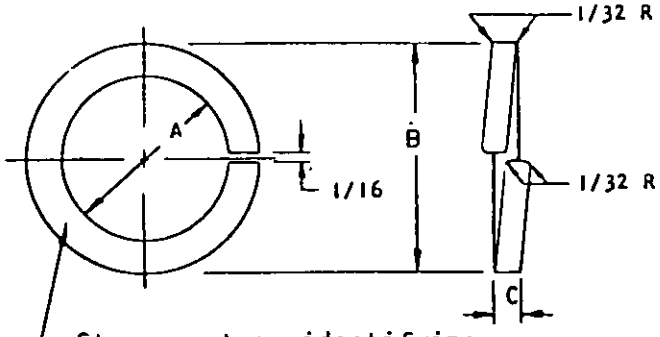
SH 9167A

FIGURE 4. Gland nut configuration. (04)

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TABLE V. Gland nut dimensions, brass.

Tube size	Part or identifying number	A DIA	B THDS	C	D	E	F DIA
1	M24235/1-019	0.609	1.125	1-9/16	7/16	1-1/4	0.984
2	M24235/1-020	.812	1.312	1-9/16	7/16	1-7/16	1.171
3	M24235/1-021	1.062	1.625	1-9/16	7/16	1-25/32	1.484
4	M24235/1-022	1.265	1.812	1-7/8	7/16	1-15/16	1.671
5	M24235/1-023	1.437	2.062	2	7/16	2-1/4	1.921
6	M24235/1-024	1.625	2.312	2-1/8	7/16	2-15/32	2.173
7	M24235/1-025	1.875	2.688	2-5/16	7/16	2-13/16	2.531
8	M24235/1-026	2.187	3.000	2-5/16	7/16	3-3/32	2.850
9	M24235/1-027	2.312	3.125	2-3/8	7/16	3-9/32	2.968



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FIGURE 5. Lockwasher configurations. (05)

TABLE VI. Lockwasher dimensions.

Tube size	Part or identifying number	A DIA	B DIA (max)	C
1	M24235/1-028	0.609	0.984	1/8
2	M24235/1-029	.812	1.171	1/8
3	M24235/1-030	1.062	1.484	3/16
4	M24235/1-031	1.265	1.671	3/16
5	M24235/1-032	1.437	1.921	1/4
6	M24235/1-033	1.625	2.173	1/4
7	M24235/1-034	1.875	2.531	1/4
8	M24235/1-035	2.187	2.850	1/4
9	M24235/1-036	2.312	2.968	1/4

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REQUIREMENTS:

1. Packing assemblies: For information only, not furnished with stuffing tube.

TABLE VII. Packing assemblies.

Tube size	Stuffing tube part or identifying number	Symbol number	Packing assembly part or identifying number	Symbol number
1	M24235/1-001 M24235/1-101	2405 HY-80 2405 HT	M24235/2-001 M24235/2-002 M24235/2-003 M24235/2-004 M24235/2-005 M24235/2-006 M24235/2-007	2405.1 2405.2 2405.3 2405.4 2405.5 2405.6 2405.7
2	M24235/1-002 M24235/1-102	2406 HY-80 2406 HT	M24235/2-008 M24235/2-009 M24235/2-010 M24235/2-011 M24235/2-012	2406.1 2406.2 2406.3 2406.5 2406.4
3	M24235/1-003 M24235/1-103	2407 HY-80 2407 HT	M24235/2-013 M24235/2-014 M24235/2-015 M24235/2-016 M24235/2-017 M24235/2-018 M24235/2-019	2407.1 2407.2 2407.5 2407.3 2407.6 2407.7 2407.4
4	M24235/1-004 M24235/1-104	2408 HY-80 2408 HT	M24235/2-030 M24235/2-031 M24235/2-032 M24235/2-033 M24235/2-034	2408.5 2408.1 2408.2 2408.3 2408.4
5	M24235/1-005 M24235/1-105	2409 HY-80 2409 HT	M24235/2-035 M24235/2-036 M24235/2-037	2409.1 2409.2 2409.3
6	M24235/1-006 M24235/1-106	2410 HY-80 2410 HT	M24235/2-038 M24235/2-039 M24235/2-040	2410.1 2410.2 2410.3
7	M24235/1-007 M24235/1-107	2411 HY-80 2411 HT	M24235/2-041 M24235/2-042 M24235/2-043 M24235/2-044 M24235/2-045	2411.5 2411.1 2411.2 2411.3 2411.4

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TABLE VII. Packing assemblies - Continued.

Tube size	Stuffing tube part or identifying number	Symbol number	Packing assembly part or identifying number	Symbol number
8	M24235/1-008 M24235/1-108	2412 HY-80 2412 HT	M24235/2-046 M24235/2-047 M24235/2-048	2412.1 2412.2 2412.3
9	M24235/1-009 M24235/1-109	2413 HY-80 2413 HT	M24235/2-049 M24235/2-050	2413.1 2413.2

2. Intended use: The stuffing tubes are intended for electric cable penetrations of pressureproof submarine bulkheads and sonar domes that are filled with water under normal operating conditions. Stuffing tubes, as shown, may be used in bulkheads of 1/2-inch maximum thickness. Bulkheads of HY-80 material shall use tubes manufactured from HY-80 grade steel; grade HT bulkheads shall use tubes of ASTM A 322, grade 8620 or ASTM A 441 steel.
3. Application: See DOD-STD-2003, section 3.
4. For special applications requiring an extended tube body exceeding 11-7/8 inches in length, the dimensional tolerance of the tube body length shall be plus or minus 1/32-inch.

TABLE VIII. Parts list.

Item no.	Description 1/	Qty 2/		Material and specification	Remarks
		1-3	4-9		
01	Stuffing tube assemblies	-		------	See figure 1 and tables I and II
02	Tube body	1		HY-80 steel, MIL-S-21952; or HT steel 3/, grade 8620 ASTM A 322; or ASTM A 441	See figure 2 and table III
03	Tube body		1	HY-80 steel, MIL-S-21952; or HT steel 3/, grade 8620 ASTM A 322; or ASTM A 441	See figure 3 and table IV

See footnotes at end of table.

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TABLE VIII. Parts list - Continued.

Item no.	Description 1/	Qty 2/ 1-3	Qty 2/ 4-9	Material and specification	Remarks
04	Gland nut	2	2	Brass, coml	See figure 4 and table V
05	Lockwasher, spring	2	2	CRES, class B, style 2, FF-W-84	See figure 5 and table VI

1/ See applicable figure and table listed in "Remarks" column for part numbers and dimensions.
2/ Quantity is for one complete assembly of each size.
3/ As specified in the ordering data.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5975-N626-1)

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